

ESTIN

Basics of image processing

S4 (DS-AI)

2025-2026

Lab 02

Exercise 1

During this part of the lab, you will write a series of functions using what you have learned in previous session. The goal is to become familiar with the various concepts of image processing and pre-processing.

With the images provided, write and show the result of a function:

- **OpenImage**: takes a single argument and returns three variables (the equivalent matrix **I**, the number of rows in the matrix **L**, and the number of columns **C**).
- **Divide**: Write a function that returns the three separate channels of an image.
- **HSV**: Write a function that returns the HSV format of an image.
- **CountPix**: takes one argument and returns one variable (**N**, the number of pixels in the image).
- **FactPix**: takes three arguments (two of which are variables you can freely choose to modify the intensities of the image) and returns one variable (**I-fact**, the modified image).
- **Func-a**: Write a function that returns as output the logarithm, exponential, square, and square root of the input image.
- **Func-m**: Write a function that returns the mean μ and the standard deviation $\sqrt{\frac{\sum (x_i - \mu)^2}{N}}$ of an image.

- **Normalize:** takes three arguments and returns one output variable (**Inorm**, the normalized image, i.e., modifying the range of gray levels in the image).
- **Inverse:** takes one argument and returns one output variable (**Inv**, the inverse image, commonly referred to as the negative mode, i.e., $Inv = \max(I) - I$).
- **CalcHist:** takes one argument and returns two output variables (**H**, a vector representing the histogram of **I**, and **b**, a vector representing the equally spaced intervals, also called bins).
- **Threshold:** takes two arguments and returns one output variable (**T**, the image obtained after thresholding, i.e., values above a specified threshold are set to 255, and the remaining pixels are set to 0).
- **Func-j:** A function defined by a series of instructions (using the functions already written) to, from the path of an image, open it, display it, show its histogram, invert it, calculate its histogram after inversion, and visualize the results of the processing.
- **Func-t:** A function defined by a series of instructions to, from the path of an image, open it, display it, show its histogram, normalize it to the range [10, 50], calculate its histogram after normalization, and visualize the results of the processing. Interpret the outcome of the visualization.
- **Func-f:** A function defined by a series of instructions to, from the path of an image, open it, display it, show its histogram, threshold it with a threshold value **s = 128**, calculate its histogram after thresholding, and visualize the results of the processing.

Exercise 2

Define the matrix **I** that represents the image below:



Where a black pixel has the value 0, a gray pixel the value 128, and a white pixel the value 255. Visualize it using `plt.imshow` and examine its histogram using the `CalcHist(I)` function generated in Part I. To create a matrix of random numbers, you can use the following `Numpy` methods:

```
import numpy as np
ran_d = np.random . rand (9 ,9)
ran_dN = np.random . randn (7 x7)
ran_dINT = np.random . randint ( low = 0, high = 8, size = (8 ,8))
```

The `rand` method in this case generates a matrix of size (9x9) with random numbers between 0 and 1. The `randn` method allows for the creation of a matrix of size (7x7) whose values follow a standard normal distribution. The `randint` method in this case generates a matrix of size (8x8) with values greater than or equal to "low" and strictly less than "high". Use these functions to generate a matrix of size 512×512 . For `randn`, test using a mean of 128 and a randomly generated standard deviation. Visualize the resulting image and its histogram. What do you observe about this histogram?